LUCKY SAH

Robotics Researcher | Mechanical Engineering | Electromechanical Prototyping

Kent, OH | lsah@kent.edu | *[ADD PHONE]* | *[ADD LINKEDIN OR PORTFOLIO URL]*

PROFESSIONAL SUMMARY

Undergraduate robotics researcher and full-stack robotics developer pursuing a degree in mechatronics engineering. Develops humanoid robotic systems from mechanical design and fabrication through hardware integration, robotics software, and simulation, using SolidWorks, Autodesk Inventor, ROS, Isaac Sim, and MuJoCo. Coauthored research papers address mechanical systems, agentic robotics, and human–robot interaction.

RESEARCH & ENGINEERING EXPERIENCE

Robotics Researcher *[ADD START DATE] – Present*

Advanced Telerobotics Research (ATR) Lab, Kent State University

- Lead mechanical development of mobile robots and humanoid robotic mechanisms, designing structures, actuator mounts, and sensor assemblies from CAD through working prototypes.
- Develop ROS interfaces for actuator control and sensor feedback; build simulation models in Isaac Sim and MuJoCo to test robot motion and hardware–software integration.
- Integrate actuators, sensors, and embedded electronics; troubleshoot mechanical and control issues during testing.

Robotics Hardware Intern (Co-op) *[ADD DATES]*

PlaiPin Inc.

- Lead mechanical CAD design for a modular AI companion robot, developing the core device and interchangeable body configurations around actuators, sensors, and onboard electronics.
- Build and refine functional prototypes, resolving assembly fit and component-packaging issues through iterative fabrication and hardware integration.

Automation and Product Development Intern *2022–2024*

Thoplo Machine Pvt. Ltd. & Baliyo Ventures Pvt. Ltd.

- Designed, fabricated, assembled, and tested mechanical systems for automated production machinery.
- Assembled industrial equipment, supported PLC wiring and lab setup, and fabricated parts using machining and fiber-reinforced molding.

EDUCATION

B.S. in Mechatronics Engineering; Minor in Computer Science *Expected 2028*

Kent State University — Kent, Ohio

PUBLICATIONS & PREPRINTS

I. S. Cardenas, M. A. Arnett, N. C. Yeo, **L. Sah**, and J.-H. Kim, "ROSClaw: An OpenClaw ROS 2 Framework for Agentic Robot Control and Interaction," *arXiv preprint*, arXiv:2603.26997, 2026.

M. Chen, **L. Sah**, A. Rivera, and D. Morgan, "Design and Experimental Evaluation of a Low-Cost Modular Mobile Robot for Human–Robot Interaction Research," *Proceedings of the Sample Robotics Research Conference*, pp. 112–119, 2026. *(fictional layout sample)*.

A. Rivera, M. Chen, **L. Sah**, and D. Morgan, "A Compact Tendon-Driven Actuation Mechanism for Expressive Humanoid Robotic Heads," *Proceedings of the Sample Symposium on Humanoid Robotics*, pp. 48–55, 2026. *(fictional layout sample)*.

SELECTED ENGINEERING PROJECTS

Mini TurtleBot — Low-Cost Modular Mobile Robot | ATR Lab

- Designed a modular mobile robot chassis and motion architecture, supporting low-cost fabrication and hardware expansion. *[ADD BUILD COST OR TEST RESULT]*

KAMI Head — Human-Robot Interaction Module

- Designed an expressive robotic head integrating actuation and embedded electronics for human–robot interaction; work contributed to a **CoRL 2025 workshop poster paper**.

TECHNICAL SKILLS

Robotics & Simulation: ROS, Isaac Sim, MuJoCo
Mechanical Design: SolidWorks, Autodesk Inventor, Fusion 360, parametric CAD, mechanism design, BOM development
Embedded & Controls: Arduino, microcontrollers, sensor integration, motor drivers, PLC fundamentals
Fabrication: FDM/resin 3D printing, CNC machining, soldering, silicone, polyurethane, and fiber-reinforced molding

ROBOTICS COMPETITIONS

World Robot Olympiad (Thailand); FIRST Global Challenge (Switzerland). *[ADD ROLES, YEARS, AND RESULTS]*